LAPORAN PRAKTIKUM

POSTTEST (3)

ALGORITMA PEMROGRAMAN LANJUT

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PROGRAM STUDI INFORMATIKA

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SAMARINDA

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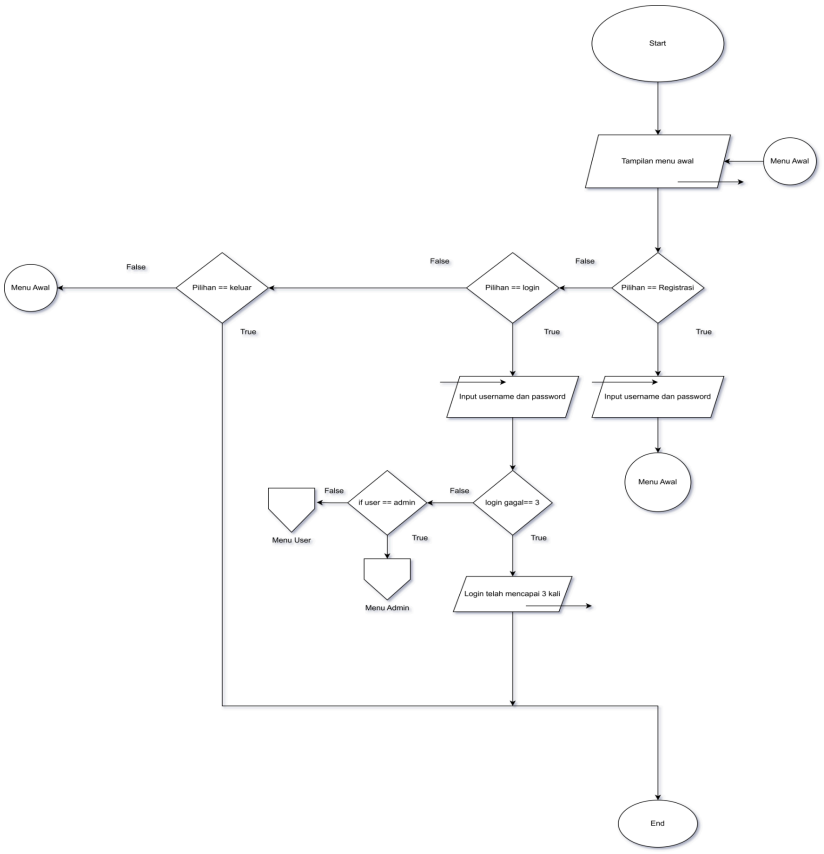
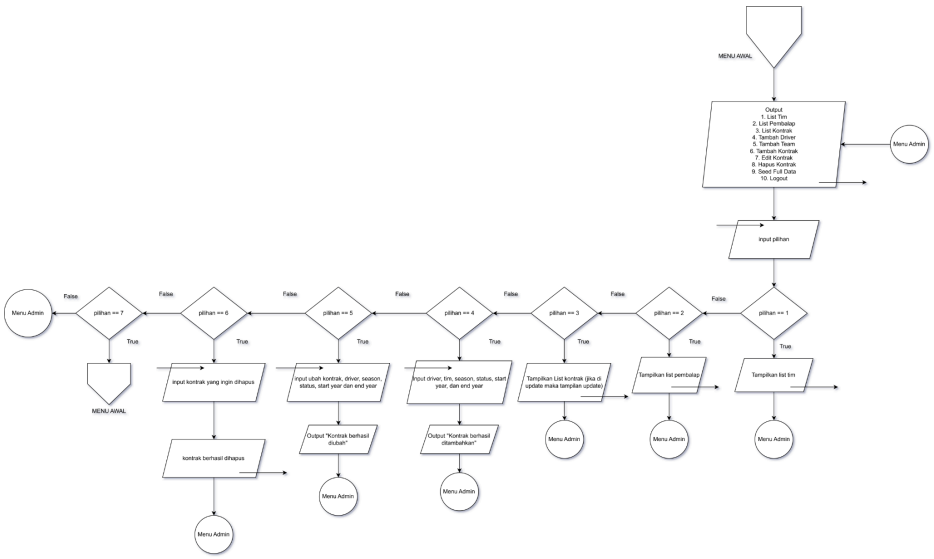
1. Flowchart

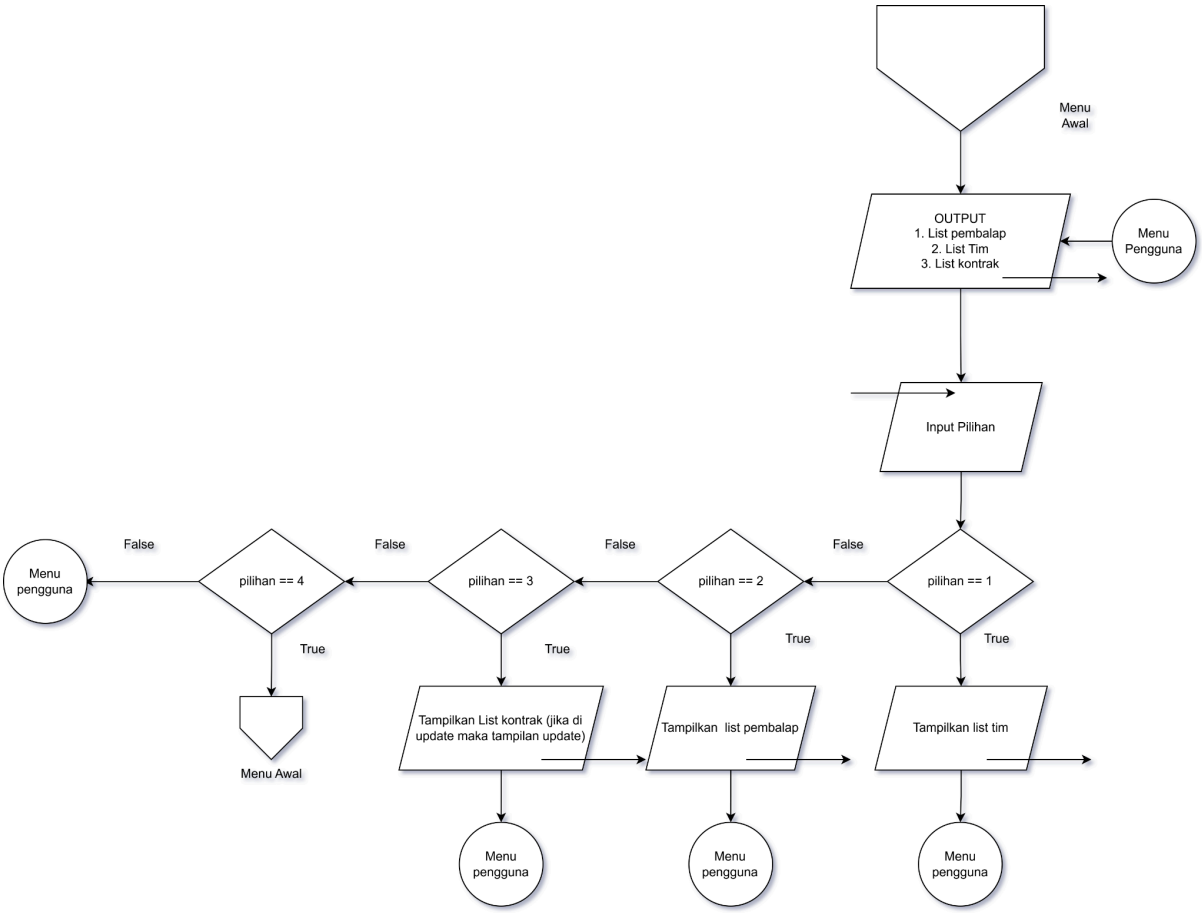
Flowchart ini membantu memvisualisasikan keputusan bercabang (pakai simbol belah ketupat) dan tindakan proses (pakai persegi panjang), sehingga mudah melihat jalur mana yang ditempuh sesuai kondisi



Gambar 1.1 Flowchart CRUD Menu awal



Gambar 1.2 Flowchart Menu Admin



Gambar 1.3 Flowchart menu Pengguna

2. Deskripsi Singkat Program

Secara keseluruhan, program ini berfungsi sebagai sistem manajemen kontrak dan informasi pembalap F1 yang sederhana, dengan kontrol akses berdasarkan role dan menu interaktif di konsol.

3. Source Code

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

struct User {
    string username;
    string password;
    bool isAdmin;
};

struct Driver {
    string name;
    string nationality;
    int birthYear;
    string status;
};

struct Team {
    string name;
    bool active;
};

struct Contract {
    int driverIndex;
    int teamIndex;
    int season;
    string status;
    int startYear;
    int endYear;
};

const int NUM_DRIVERS = 22;
Driver drivers[NUM_DRIVERS] = {
```

```
    {"Charles Leclerc","Monaco",1997,"Active"},
    {"Lewis Hamilton","UK",1985,"Active"},
    {"George Russel","UK",1998,"Active"},
    {"Kimi Antonelli","Italy",2004,"Active"},
    {"Lando Norris","UK",1999,"Active"},
    {"Oscar Piastri","Australia",2001,"Active"},
    {"Max Verstappen","Netherlands",1997,"Active"},
    {"Isack Hadjar","France",2003,"Active"},
    {"Fernando Alonso","Spain",1981,"Active"},
    {"Lance Stroll","Canada",2000,"Active"},
    {"Pierre Gasly","France",1996,"Active"},
    {"Franco Colapinto","Argentina",2003,"Active"},
    {"Carlos Sainz","Spain",2001,"Active"},
    {"Alex Albon","Thailand",1996,"Active"},
    {"Esteban Ocon","France",1996,"Active"},
    {"Oliver Bearman","UK",2005,"Active"},
    {"Liam Lawson","NZ",2001,"Active"},
    {"Arvid Lindblad","Sweden",2004,"Active"},
    {"Nico Hulkenberg","Germany",1987,"Active"},
    {"Gabriel Bortoleto","Brazil",2004,"Active"},
    {"Sergio Perez","Mexico",1990,"Active"},
    {"Valteri Bottas","Finland",1989,"Active"}
};

const int NUM_TEAMS = 11;
Team teams[NUM_TEAMS] = {
    {"Scuderia Ferrari HP",true},
    {"Mercedes AMG-Petronas",true},
    {"Oracle RedBull Racing",true},
    {"McLaren Mastercard",true},
    {"Aston Martin Aramco",true},
    {"BWT Alpine",true},
    {"Atlassian Williams",true},
    {"TGR Haas",true},
    {"VCAB RB",true},
    {"Audi Revolut",true},
    {"Cadillac",true}
};

Contract contracts[100];
int contractsCount = 0;

User users[50];
int usersCount = 0;
int currentUserIndex = -1;

void listDrivers() {
    cout << "\n=== Daftar Pembalap ===\n";
```

```
    cout << left << setw(5) << "No" << setw(25) << "Nama" << setw(15) << 
"Kebangsaan"
         << setw(5) << "Umur" << setw(10) << "Status" << "\n";
    cout << string(65,'-') << "\n";
    for(int i=0;i<NUM_DRIVERS;i++){
        int age = 2026 - drivers[i].birthYear;
        cout << setw(5) << (i+1)
             << setw(25) << drivers[i].name
             << setw(15) << drivers[i].nationality
             << setw(5) << age
             << setw(10) << drivers[i].status << "\n";
    }
}

void listTeams() {
    cout << "\n=== Daftar Tim ===\n";
    cout << left << setw(5) << "No" << setw(30) << "Nama" << setw(10) << 
"Status" << "\n";
    cout << string(45,'-') << "\n";
    for(int i=0;i<NUM_TEAMS;i++){
        cout << setw(5) << (i+1)
             << setw(30) << teams[i].name
             << setw(10) << (teams[i].active?"Active":"Inactive") << "\n";
    }
}

void listContracts() {
    if(contractsCount==0){ cout << "Belum ada kontrak.\n"; return; }
    cout << "\n=== Daftar Kontrak ===\n";
    cout << left << setw(5) << "No" << setw(25) << "Driver" << setw(25) << 
"Team"
         << setw(8) << "Season" << setw(10) << "Status"
         << setw(10) << "Start" << setw(10) << "End" << "\n";
    cout << string(93,'-') << "\n";
    for(int i=0;i<contractsCount;i++){
        cout << setw(5) << (i+1)
             << setw(25) << drivers[contracts[i].driverIndex].name
             << setw(25) << teams[contracts[i].teamIndex].name
             << setw(8) << contracts[i].season
             << setw(10) << contracts[i].status
             << setw(10) << contracts[i].startYear
             << setw(10) << contracts[i].endYear << "\n";
    }
}

void registerUser(bool isAdmin) {
    cin.ignore();
    string username, password;
```

```
    cout << "Masukkan username baru: ";
    getline(cin, username);
    cout << "Masukkan password baru: ";
    getline(cin, password);
    users[usersCount++] = User{username,password,isAdmin};
    cout << "Registrasi berhasil.\n";
}

bool loginUser() {
    cin.ignore();
    string username,password;
    int attempts = 0;
    while(attempts < 3){
        cout << "Username: "; getline(cin, username);
        cout << "Password: "; getline(cin, password);
        for(int i=0;i<usersCount;i++){
            if(users[i].username==username && users[i].password==password){
                currentUserIndex = i;
                cout << "Login berhasil! Selamat datang, " << username <<
"\n";
                return true;
            }
        }
        attempts++;
        cout << "Login gagal ("<<attempts<<"/3)\n";
        if(attempts==3){
            cout << "Anda salah 3 kali. Program berhenti.\n";
            exit(0);
        }
    }
    return false;
}

void addContract() {
    listDrivers();
    cout << "Pilih nomor driver: "; int d; cin >> d; cin.ignore(); d--;
    listTeams();
    cout << "Pilih nomor team: "; int t; cin >> t; cin.ignore(); t--;
    cout << "Season: "; int s; cin >> s; cin.ignore();
    cout << "Status (Rumor/Resmi): "; string status; getline(cin,status);
    cout << "Start Year: "; int start; cin >> start; cin.ignore();
    cout << "End Year: "; int end; cin >> end; cin.ignore();
    contracts[contractsCount++] = Contract{d,t,s,status,start,end};
    cout << "Kontrak berhasil ditambahkan.\n";
}

void editContract() {
    if(contractsCount==0){ cout<<"Belum ada kontrak.\n"; return;}
```

```
    listContracts();
    cout << "Pilih nomor kontrak untuk diedit: "; int idx; cin >> idx;
cin.ignore(); idx--;
    if(idx<0 || idx>=contractsCount){ cout<<"Nomor salah\n"; return;}

    listDrivers();
    cout << "Driver baru (0=tidak diubah): "; int d; cin >> d; cin.ignore();
    if(d>0 && d<=NUM_DRIVERS) contracts[idx].driverIndex=d-1;

    listTeams();
    cout << "Team baru (0=tidak diubah): "; int t; cin >> t; cin.ignore();
    if(t>0 && t<=NUM_TEAMS) contracts[idx].teamIndex=t-1;

    cout << "Season baru (0=tidak diubah): "; int s; cin >> s; cin.ignore();
    if(s>0) contracts[idx].season=s;

    cout << "Status baru (kosong=tidak diubah): "; string status;
getline(cin,status);
    if(!status.empty()) contracts[idx].status=status;

    cout << "Start Year baru (0=tidak diubah): "; int start; cin >> start;
cin.ignore();
    if(start>0) contracts[idx].startYear=start;

    cout << "End Year baru (0=tidak diubah): "; int end; cin >> end;
cin.ignore();
    if(end>0) contracts[idx].endYear=end;

    cout << "Kontrak berhasil diperbarui.\n";
}

void deleteContract() {
    if(contractsCount==0){ cout<<"Belum ada kontrak.\n"; return;}
    listContracts();
    cout << "Pilih nomor kontrak untuk dihapus: "; int idx; cin >> idx;
cin.ignore(); idx--;
    if(idx<0 || idx>=contractsCount){ cout<<"Nomor salah\n"; return;}
    for(int i=idx;i<contractsCount-1;i++) contracts[i]=contracts[i+1];
    contractsCount--;
    cout << "Kontrak berhasil dihapus.\n";
}

void adminMenu() {
    while(true){
        cout << "\n=== MENU ADMIN ===\n"
             << "1. List Tim\n2. List Pembalap\n3. List Kontrak\n"
             << "4. Tambah Kontrak\n5. Edit Kontrak\n6. Hapus Kontrak\n7.
Logout\nPilihan: ";
```

```
        int ch; cin >> ch; cin.ignore();
        switch(ch){
            case 1: listTeams(); break;
            case 2: listDrivers(); break;
            case 3: listContracts(); break;
            case 4: addContract(); break;
            case 5: editContract(); break;
            case 6: deleteContract(); break;
            case 7: return;
            default: cout << "Pilihan salah, silakan coba lagi.\n";
        }
    }
}

void userMenu() {
    while(true){
        cout << "\n=== MENU PENGGUNA ===\n"
             << "1. List Tim\n2. List Pembalap\n3. List Kontrak\n4.
Logout\nPilihan: ";
        int ch; cin >> ch; cin.ignore();
        switch(ch){
            case 1: listTeams(); break;
            case 2: listDrivers(); break;
            case 3: listContracts(); break;
            case 4: return;
            default: cout << "Pilihan salah, silakan coba lagi.\n";
        }
    }
}

int main() {
    users[usersCount++] = User{"Attol","2509106103",true};

    while(true){
        cout << "\n=== SISTEM F1 TRANSFER ===\n1. Register Pengguna\n2.
Login\n3. Keluar\nPilihan: ";
        int ch; cin >> ch; cin.ignore();
        switch(ch){
            case 1: {
                cout << "Apakah admin? (y/n): "; string resp; getline(cin,
resp);
                bool isAdmin = (resp=="y" || resp=="Y");
                registerUser(isAdmin);
                break;
            }
            case 2:
                if(loginUser()){
                    if(users[currentUserIndex].isAdmin) adminMenu();
```

```
                    else userMenu();
                }
                break;
            case 3: return 0;
            default: cout << "Pilihan salah, silakan coba lagi.\n";
        }
    }
}
```

4. Hasil Output

```
=== SISTEM F1 TRANSFER ===
1. Register Pengguna
2. Login
3. Keluar
Pilihan: 1
Apakah admin? (y/n): n

Masukkan username baru: a
Masukkan password baru: a
Registrasi berhasil.

=== SISTEM F1 TRANSFER ===
1. Register Pengguna
2. Login
3. Keluar
Pilihan: 2

Username: Attol
Password: 2509106103
Login berhasil! Selamat datang, Attol
```

Gambar 4.1 Hasil Output Menu Awal

```
=== MENU ADMIN ===
1. List Tim
2. List Pembalap
3. List Kontrak
4. Tambah Kontrak
5. Edit Kontrak
6. Hapus Kontrak
7. Logout
Pilihan:
```

Gambar 4.2 Hasil Output Login Sebagai Admin

```
=== MENU ADMIN ===
1. List Tim
2. List Pembalap
3. List Kontrak
4. Tambah Kontrak
5. Edit Kontrak
6. Hapus Kontrak
7. Logout
Pilihan: 1

=== Daftar Tim ===
No   Nama                        Status
-----------------------------------------
1    Scuderia Ferrari HP         Active
2    Mercedes AMG-Petronas       Active
3    Oracle RedBull Racing       Active
4    McLaren Mastercard          Active
5    Aston Martin Aramco         Active
6    BWT Alpine                  Active
7    Atlassian Williams          Active
8    TGR Haas                    Active
9    VCAB RB                     Active
10   Audi Revolut                Active
11   Cadillac                    Active
```

Gambar 4.3 Hasil Output List Tim

```
Pilihan: 2

=== Daftar Pembalap ===
No   Nama                    Kebangsaan     Umur Status
----------------------------------------------------------------
1    Charles Leclerc         Monaco         29   Active
2    Lewis Hamilton          UK             41   Active
3    George Russel           UK             28   Active
4    Kimi Antonelli          Italy          22   Active
5    Lando Norris            UK             27   Active
6    Oscar Piastri           Australia      25   Active
7    Max Verstappen          Netherlands    29   Active
8    Isack Hadjar            France         23   Active
9    Fernando Alonso         Spain          45   Active
10   Lance Stroll            Canada         26   Active
11   Pierre Gasly            France         30   Active
12   Franco Colapinto        Argentina      23   Active
13   Carlos Sainz            Spain          25   Active
14   Alex Albon              Thailand       30   Active
15   Esteban Ocon            France         30   Active
16   Oliver Bearman          UK             21   Active
17   Liam Lawson             NZ             25   Active
18   Arvid Lindblad          Sweden         22   Active
19   Nico Hulkenberg         Germany        39   Active
20   Gabriel Bortoleto       Brazil         22   Active
21   Sergio Perez            Mexico         36   Active
22   Valteri Bottas          Finland        37   Active
```

Gambar 4.4 Hasil Output List pembalap

```
14   Alex Albon              Thailand       30   Active
15   Esteban Ocon            France         30   Active
16   Oliver Bearman          UK             21   Active
17   Liam Lawson             NZ             25   Active
18   Arvid Lindblad          Sweden         22   Active
19   Nico Hulkenberg         Germany        39   Active
20   Gabriel Bortoleto       Brazil         22   Active
21   Sergio Perez            Mexico         36   Active
22   Valteri Bottas          Finland        37   Active
Pilih nomor driver: 14

=== Daftar Tim ===
No   Nama                          Status
---------------------------------------------
1    Scuderia Ferrari HP           Active
2    Mercedes AMG-Petronas         Active
3    Oracle RedBull Racing         Active
4    McLaren Mastercard            Active
5    Aston Martin Aramco           Active
6    BWT Alpine                    Active
7    Atlassian Williams            Active
8    TGR Haas                      Active
9    VCAB RB                       Active
10   Audi Revolut                  Active
11   Cadillac                      Active
Pilih nomor team: 1
Season: 2027
Status (Rumor/Resmi): Rumor
Start Year: 2027
End Year: 2030
Kontrak berhasil ditambahkan.
```

Gambar 4.5 Output Create

```
=== Daftar Kontrak ===
No   Driver                  Team                    Season  Status    Start     End
-------------------------------------------------------------------------------------------
1    Alex Albon              Scuderia Ferrari HP     2027    Rumor     2027      2030
Pilih nomor kontrak untuk dihapus: 1
Kontrak berhasil dihapus.

=== MENU ADMIN ===
1. List Tim
2. List Pembalap
3. List Kontrak
4. Tambah Kontrak
5. Edit Kontrak
6. Hapus Kontrak
7. Logout
Pilihan: 3
Belum ada kontrak.
```

4.6 Output Delete

```
Team baru (0=tidak diubah): 0
Season baru (0=tidak diubah): 0
Status baru (kosong=tidak diubah):
Start Year baru (0=tidak diubah): 0
End Year baru (0=tidak diubah): 0
Kontrak berhasil diperbarui.

=== MENU ADMIN ===
1. List Tim
2. List Pembalap
3. List Kontrak
4. Tambah Kontrak
5. Edit Kontrak
6. Hapus Kontrak
7. Logout
Pilihan: 3

=== Daftar Kontrak ===
No   Driver                  Team                    Season  Status    Start     End
-----------------------------------------------------------------------------------------
1    Max Verstappen          McLaren Mastercard      2030    rumor     2030      2033
```

Gambar 4.7 Hasil Output Edit kontrak

```
=== MENU PENGGUNA ===
1. List Tim
2. List Pembalap
3. List Kontrak
4. Logout
Pilihan: 3

=== Daftar Kontrak ===
No   Driver                  Team                    Season  Status    Start     End
-----------------------------------------------------------------------------------------
1    Max Verstappen          McLaren Mastercard      2030    rumor     2030      2033

=== MENU PENGGUNA ===
1. List Tim
2. List Pembalap
3. List Kontrak
4. Logout
Pilihan:
```

Gambar 4.8 Hasil Output Menu Pengguna

5. Langkah-langkah GIT

```
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git add .
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git commit -m "KOMAT KAMIT"
[main aede32e] KOMAT KAMIT
 3 files changed, 233 insertions(+)
 create mode 100644 post-test/post-test-7/2509106103-KhayzanDwitraAttala-PT-7 (1).pdf
 create mode 100644 post-test/post-test-7/2509106103-PT-7.drawio.png
 create mode 100644 post-test/post-test-7/2509106103_KhayzanDwitraAttala-PT-7.py
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 868.02 KiB | 27.13 MiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Khayzannn/praktikum-apd.git
   609b85f..aede32e  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7>
```

Gambar 5.1 Langkah - Langkah Git

5.1 GIT Add

Git add adalah perintah diGit yang digunakan untuk menambahkan perubahan pada berkas di direktori kerja ke area staging (atau indeks), mempersiapkannya untuk dimasukkan ke dalam commit berikutnya.

5.2 GIT Commit

Git commit adalah perintah di sistem kontrol versi Git untuk menyimpan (melakukan "commit") serangkaian perubahan yang telah dipilih (di-staging) ke dalam repositori lokal Anda, membuat sebuah "snapshot" atau titik pemeriksaan pada riwayat proyek dengan deskripsi perubahan.

5.3 GIT Push

Git push adalah perintah dalam Git yang digunakan untuk mengunggah (mengirimkan) perubahan yang telah dicatat di repositori lokal Anda ke repositori jarak jauh (seperti GitHub), sehingga memperbarui cabang tersebut dan membagikan pekerjaan dengan tim atau menyinkronkan dengan versi pusat.